

Rhizopus oryzae (version 3; http://www.broadinstitute.org/annotation/genome/rhizopus_oryzae, last accessed December 13, 2013), the filasterean *Capsaspora owczarzaki* (version 2; http://www.broadinstitute.org/annotation/genome/multicellularity_project/, last accessed December 13, 2013), and the sea anemone *Nematostella vectensis* (version 1; <http://genome.jgi-psf.org/Nemve1/>, last accessed December 13, 2013).

Protein Domain Architecture and TK Classification

Protein domains composing the ichthyosporean and corallochytrean TKs were annotated by the search of SMART (<http://smart.embl-heidelberg.de>, last accessed December 13, 2013) and Pfam (<http://pfam.sanger.ac.uk/>, last accessed December 13, 2013) databases, which were then subjected to manual inspections of the result alignments. Sequences that were not mapped to any known domain were further searched for in the public databases and in other ichthyosporean TKs, and two domains (I1 and I3 domains) were newly annotated. TM segments and signal peptides of RTKs were manually predicted. For the detailed classification of CTK families, their N-terminal myristoylation sites were predicted by the Myristoylator program (Bologna et al. 2004), which provides the prediction confidence by the score S. We classified TKs into families based on their domain architectures and phylogenetic positions as in (Suga et al. 2012). TKs without clear domain architectural information or an apparent affiliation to a specific family supported by the phylogeny were provisionally classified into IchRTK20 (for ichthyosporean genes) or CorRTK4 (for corallochytrean genes) families.

Network Analysis of Domain Architecture

Domain architectures of proteins containing the four pTyr signaling domains (TK catalytic, PTP catalytic, SH2, and PTB domains) were manually examined. The domain combinations occurring in these proteins were represented by a network drawn by Cytoscape (<http://www.cytoscape.org/>, last accessed December 13, 2013). In the network, co-occurring two domains are represented by two circles directly connected by a line. Duplicated domains in a protein were not counted.

Phylogenetic Tree Inference

Phylogenetic trees were inferred from the kinase (catalytic) domain sequences (225 amino acid positions) by the Maximum Likelihood (ML) method, using the LG- Γ model with four rate categories implemented in RAxML version 7.2.8 (Stamatakis 2006). The alignment is in supplementary fig. S10, Supplementary Material online. The obtained topology was further optimized by the genetic algorithm-based heuristic approach (Katoh et al. 2001; Suga, Schmid, et al. 2008; Suga et al. 2012), in which all the possible topologies generated both by crossing over all trees in the population and by the subtree pruning and regrafting (SPR) and tree bisection and reconnection (TBR) algorithms started from the best tree were evaluated. To avoid the long-branch attraction artifact, Awh_RS39176 of *Abeoforma* and Apa_RS28315.0.1 of *Amoebidium* were excluded from phylogenetic analyses.